भारतीय प्रौद्योगिकी
संस्थान जम्मू
INDIAN INSTITUTE OF
TECHNOLOGY JAMMU

Class Test-2

Engineering Mechanics ME-1-01(TO)

Date: 15/10/2025 | **Duration:50Min**
Maximum Marks: 20 | **Weightage: 10%**

Note: i) Attempt All questions. ii) Use of scientific calculator is allowed

Q.1. The mechanism is used to weigh mail A package placed at mail (Fig. 1). A package placed at A causes the weighted pointer to rotate through an angle α. Neglect the weights of the members except for the the members except for the counterweight at B, which has a mass of 4 kg. If $\alpha = 20°$, what is the mass of the package at the package at A? **(04)**

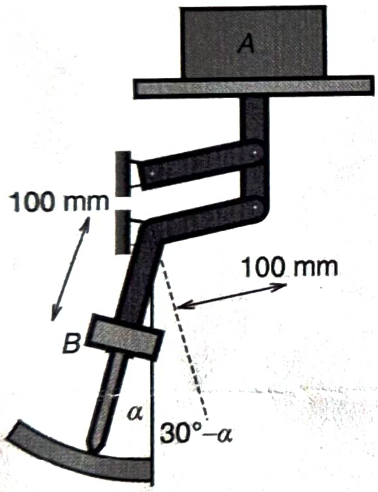

Fig. 1

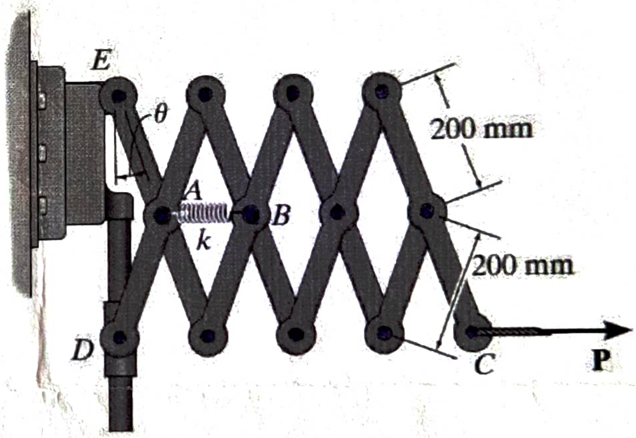

Fig. 2

K= 15kN/m
P =600N.

Q.2. The "Nuremberg scissors" (see Fig 2) is subjected to a horizontal force of Determine the angle for equilibrium. The spring has a stiffness of and is unstretched when $\theta = 15°$. **(08)**

Q.3. Locate the centroid of the shaded region of Fig 3. **(03)**

Q.4. Determine the moment of inertia of the composite area (Fig 4) about the x axis. **(04)**

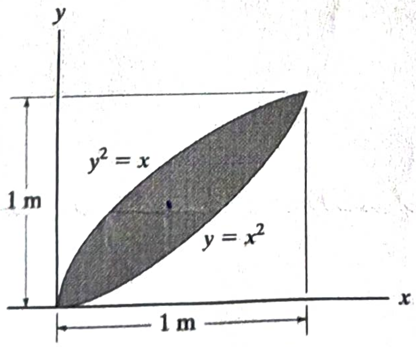

Fig 3

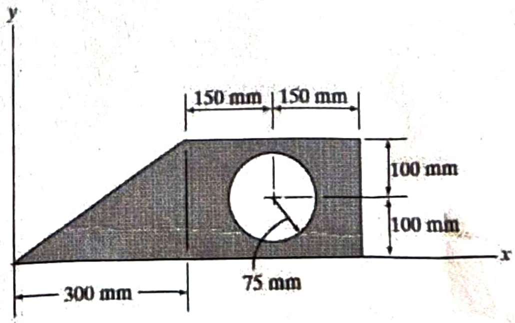

Fig 4

game
[redacted] of Luck

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Q.1(a) In the design of the robot to insert the small cylindrical part into a close-fitting circular hole, the robot arm must exert a 90-N force P on the part parallel to the axis of the hole as shown. Determine the components of the force which the part exerts on the robot along axes (a) parallel and perpendicular to the arm AB, and (b) parallel an perpendicular to the arm BC. [05 Marks]

Q1 (b)Elements of the lower arm are shown in the figure. The mass of the forearm is 2.3 kg with mass center at G. Determine the combined moment about the elbow pivot O of the weights of the forearm and the 3.6-kg homogeneous sphere. What must the biceps tension force be so that the overall moment about O is zero? [05 Marks]

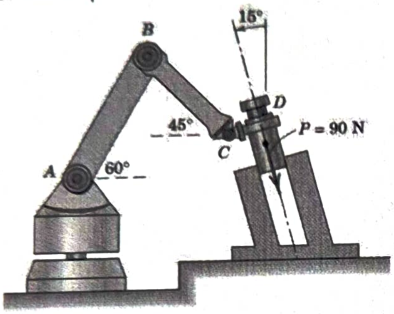

Fig 1 (a)

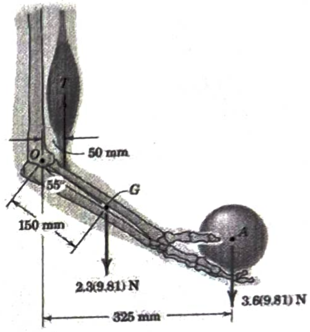

Fig 1 (b)

Q 2(a) For a coplanar, non-concurrent force system shown in Fig 2(a). determine magnitude, direction and position with reference to point A of resultant force. [05 Marks]

Q 2(b) Calculate reactions at support due to applied load on the beam as shown in Fig 2 (b).

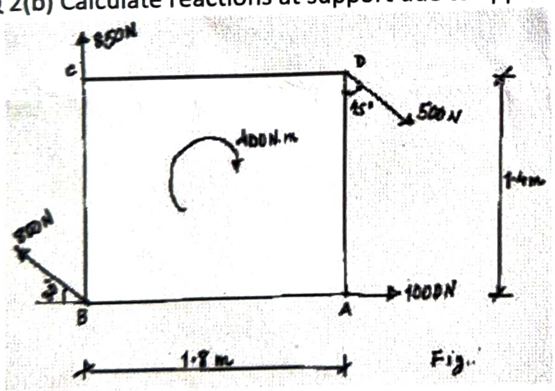

Fig 2 (a)

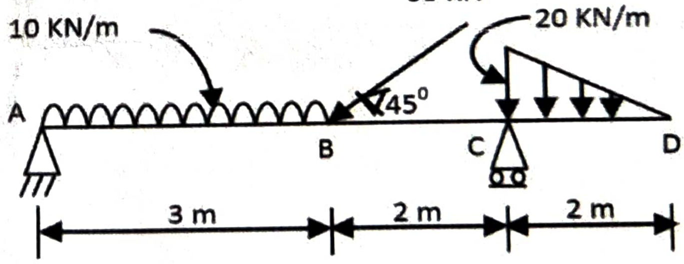

Fig 2 (b)

Q 3(a) The jib crane (Fig 3(a)) is designed for a maximum capacity of 10 kN, and its uniform I-beam has a mass of 200 kg. (a) Plot the magnitude R of the force on the pin at A as a function of x through its operating range of $x = 0.2$ m to $x = 3.8$ m. On the same set of axes, plot the x- and y-components of the pin reaction at A. (b) Determine the minimum value of R and the corresponding value of x. (c) For what value of R should the pin at A be designed? [5 Marks]

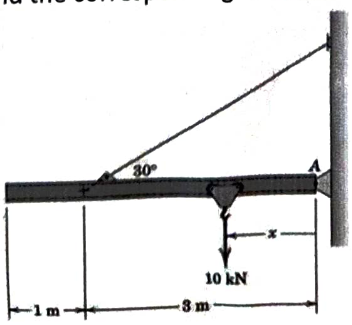

Fig 3 (a)

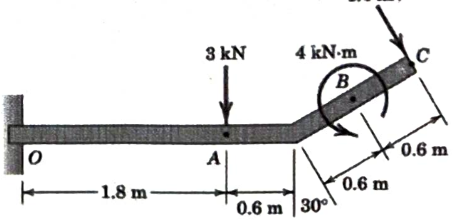

Fig 3 (b)

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Q 3(b) The uniform beam (Fig 3 (b)) has a mass of 50 kg per meter of length. Compute the reactions at the support O. The force loads shown lie in a vertical plane. [5 Marks]

Q. 4 Determine the forces in the members BC,BE and AE of the truss shown in figure and indicate the magnitude and nature of the forces on the diagram of the truss. All inclined members are at 60° to horizontal and length of each member is 2m.(Using Method of Sections) [10 Marks]

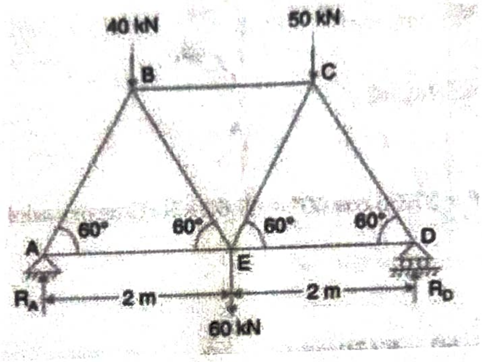

Fig 4

Q 5 (a) A 500-kg log (Fig 5 (a)) is being steadily pulled up the incline by means of the cable attached to the winch on the truck. If the coefficient of kinetic friction is 0.80 between the log and the incline and 0.50 between the cable and rock, determine the tension **T** which must be developed by the winch. [5 Marks]

Q 5 (b) A compressive force of 600 N (Fig 5 (b)) is to be applied to the two boards in the grip of the C-clamp. The threaded screw has a mean diameter of 10 mm and advances 2.5 mm per turn. The coefficient of static friction is 0.20. Determine the force F which must be applied normal to the handle at C in order to (a) tighten and (b) loosen the clamp. Neglect friction at point A. [5 Marks]

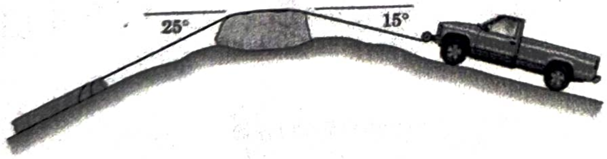

Fig 5 (a)

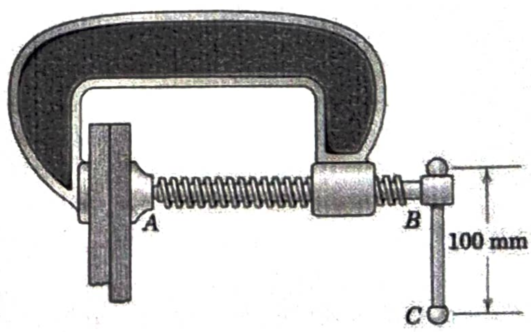

Fig 5 (b)

Q 6 (a) The cross section of a trap door hinged at A and having a mass m and a center of mass at G is shown in the figure. The spring is compressed by the rod which is pinned to the lower end of the door and which passes through the swivel block at B. When $\theta = 0$, the spring is undeformed. Show that with the proper stiffness k of the spring, the door will be in equilibrium for any angle θ.

Q 6 (b) For the mechanism shown, the spring of stiffness k has an unstretched length of essentially zero, and the larger link has a mass m with mass center at B. The mass of the smaller link is negligible. Determine the equilibrium angle & for a given downward force P.

Q7 (a) A 5 kNm torque is applied to the vertical shaft CD shown in Fig. 7(a), which allows the 10-kg gear A to turn freely about CE. Assuming that gear A starts from rest, determine the angular velocity of CD after it has turned two revolutions. Neglect the mass of shaft CD and axle CE and assume that gear A can be approximated by a thin disk. Gear B is fixed. [5 Marks]

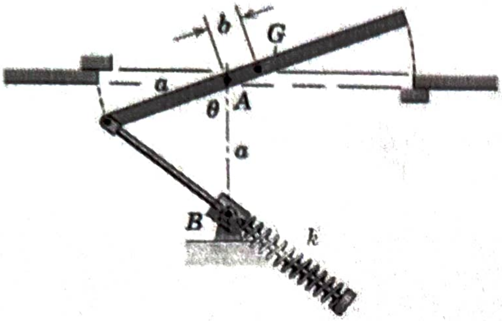

Fig 6 (a)

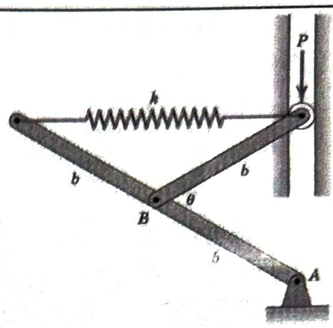

Fig 6 (b)

Q7(b) If the frame rotates with a constant angular velocity of $\omega_P = \{-10\mathbf{k}\}$ rad/sec and the horizontal gear B rotate with a constant angular velocity of $\omega_B = \{5\mathbf{k}\}$ rad/sec, determine the angular velocity and angular acceleration of the bevel gear A (as shown in Fig 7 (b)). The **k** represents unit vector direction in **z**. [5 Marks]

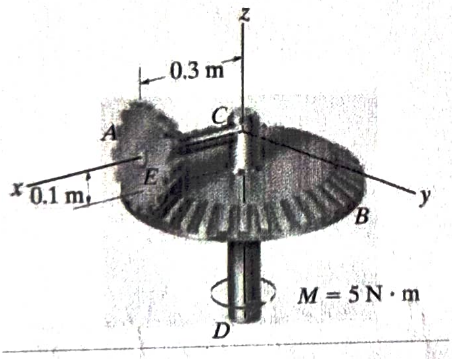

Fig 7(a)

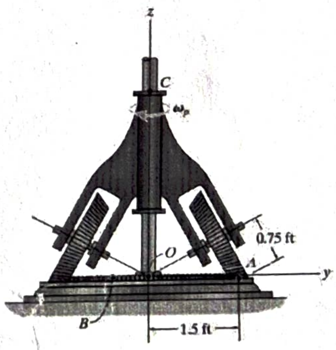

Fig 7 (b)

Q8 (a) At the instant shown in Fig 8 (a), the shaft and plate rotates with an angular velocity of $\omega = 14$ rad/sec and angular acceleration of $\alpha = 7$ rad/sec^2. Determine the velocity and acceleration of point D located on the corner of the plate at this instant. Express the result in Cartesian vector form. [5 Marks]

Q8 (b) The smooth particle has a mass of 80g (see Fig 8 (b)). It is attached to an elastic cord extending from O to P and due to the slotted arm guide moves along the horizontal circular path $r = (0.8 \sin\theta)$ m. If the cord has a stiffness of k = 30 N/m and an unstretched length of 0.25 m, determine the force of the guide on the particle when θ=60°. The guide has a constant angular velocity $\theta = 5$ rad/s. [5 Marks]

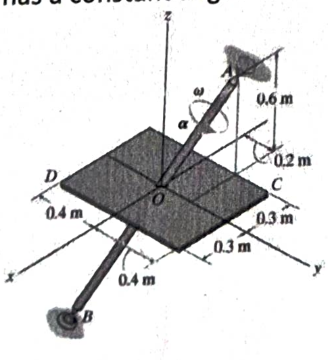

Fig 8(a)

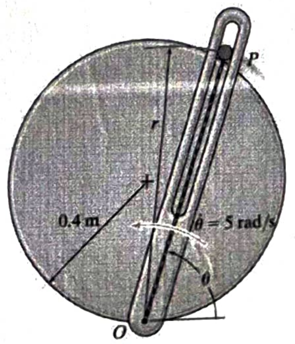

Fig 8 (b)

Q9) The planetary gear (Fig 9) set of an automatic transmission consists of three planet gears A, B, and C, mounted on carrier D, and meshed with sun gear E and ring gear F. By controlling which gear of the planetary set rotates and which gear receives the engine's power, the automatic transmission can alter a car's speed and direction. If the ring gear is held stationary and the carrier is rotating with a clockwise angular velocity of $\omega = 20$ rad/sec, determine the angular velocity of the planet gears and the sun gear. The radii of the planet gears and the sun gear are 45 mm and 75 mm, respectively. [10 Marks]

Q 10Calculate the moment of inertia and radius of gyration about the x-axis for the shaded area shown in Fig 10. Wherever possible, make expedient use of tabulated moments of inertia. [10 Marks]

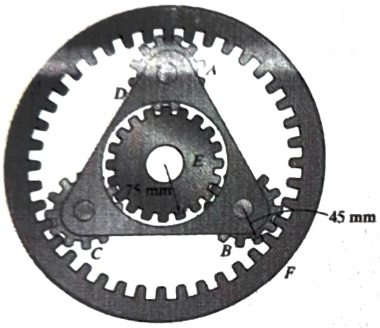

Fig 9

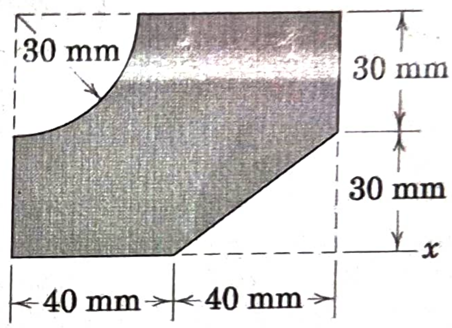

Fig 10

60×80
4800+